

Supplement 1. Numerical Simulation Parameters

To ensure the integrity of the 100-cycle stress test, the numerical integration of the Lindblad Master Equation was performed under the following rigorously defined parameters:

Basis Dimension (N): 100 Fock states. Convergence was verified by ensuring that the occupancy of the upper boundary ($n = 100$) remained below 10^{-12} , eliminating any artificial boundary effects.

Operational Frequency (Ω): 1.0 (normalized). All time and energy scales are expressed relative to the 50 GHz carrier frequency.

Non-adiabaticity Parameter (η): 1.2. This represents a strong excitation regime, where $\eta(t) \equiv |\dot{\Omega}(t)|/\Omega^2(t)$, with $\Omega(t)$ denoting the instantaneous mode frequency. This adopts the self-referencing limit $\omega \rightarrow \Omega$ discussed in Section 1, appropriate for a numerical simulation with no separate external probe.

Anharmonicity / Nonlinear regulator (U): 0.12. This value is the numerically identified stability threshold (Fig. 3); it is a normalized model parameter and is not derived from a specific physical anisotropy constant (see Section 5.1).

Coupling Strength $\mathcal{G}(t)$: $\eta\Omega/4 = 0.3$. This defines the magnitude of the parametric drive during the non-adiabatic transition.

Damping Parameter (α): 0.001 (Target) and 0.15 (Current benchmark).

Integration Time: $t_{\max} = 100 \times (2\pi/\Omega) \approx 628.32 \Omega^{-1}$, covering 100 full operational cycles.

The simulation of the Fock-state occupancy (Figures 1 and 2) was performed using the QuTip mesolve engine with an adaptive Adams integrator. This approach was specifically chosen to validate the spectral localization within the truncated basis. However, for the high-resolution phase-coherence and thermodynamic verifications (Figures 4–9), we transitioned to a custom-written Python RK4 integrator, as detailed in Supplement 2.

For the density-matrix evolution and Fock-state occupancy analysis, simulations were benchmarked at $N = 40$ and $N = 100$. Convergence was confirmed as the occupancy of the highest Fock state $|N - 1\rangle$ remained below 10^{-9} , ensuring that the Hilbert space truncation does not induce numerical reflection. The Adams/BDF method was used with an absolute tolerance of

10^{-12} and a relative tolerance of 10^{-10} . A convergence verification demonstrated that a step size of 1 fs ensures numerical integrity, with further reduction to 0.5 fs yielding a deviation in the final occupancy $\langle |v|^2 \rangle$ of less than 0.01%. The numerical integrity was monitored via the preservation of the bosonic commutation relations and the stability of the Euclidean manifold $|u|^2 + |v|^2 \approx 1 + 2 \langle |v|^2 \rangle_{\text{stable}}$ in the presence of the nonlinear regulator U .

Simulations were executed on an NVIDIA A100 GPU cluster (for Monte-Carlo batches) and high-performance CPU nodes. The total computational budget for the 1000-cycle stability map exceeded 10^{12} floating-point operations, ensuring statistically significant results for the robustness verification.

Supplement 2. Methods and Computational Protocol

To validate the theoretical framework of non-adiabatic magnon dynamics, we performed a series of numerical verifications focusing on the evolution of Bogoliubov coefficients and phase stability. The following methodology ensures the physical reproducibility of the results presented in Section 6.

1. Computational Framework and Approximations

Nonlinear Bogoliubov Equations: Instead of a truncated Fock basis, we solved the time-dependent system for complex amplitudes $u(t)$ and $v(t)$, governed by the Hamiltonian with a self-consistent nonlinear stabilizer term:

$$\hat{H}(t) = \Omega(t) \hat{A}^\dagger \hat{A} + U (\hat{A}^\dagger \hat{A})^2 + \mathcal{G}(t) (\hat{A}^{\dagger 2} + \hat{A}^2)$$

Effective Anharmonicity (U): The parameter U represents the lumped contribution of 4-magnon scattering processes and magnetocrystalline anisotropy. It serves as a frequency detuning mechanism that prevents bosonic uncontrolled occupation growth.

Approximation Limits: We assumed a single-mode approximation for the primary computational cell (20 x 20 x 10 nm), where the spatial non-uniformity is negligible compared to the wavelength of the excited exchange magnons.

2. Numerical Integration Protocol

Step Size and Precision: Simulations were conducted using a custom-written Python RK4 integrator with a fixed time step of $\Delta t = 1$ fs (10^{-15} s). This resolution is critical to resolve nonlinear phase jitter and avoid aliasing at high values of U (up to $U = 5.0$).

Stability Condition: To ensure convergence, the localized bounded regime was monitored. In the stabilized regime ($U > 0.1$), the system remains locked within the logical subspace, effectively emulating a two-level Fermi-system. While the initial Fock-space analysis utilized the QuTip library, the extended precision verification required to resolve non-linear phase jitter and thermodynamic stability was conducted using a dedicated RK4 scheme. This custom implementation allows for a fixed time step ($\Delta t = 1$ fs), which is essential to avoid numerical dissipation over 1000+ computational cycles. Unlike adaptive step-size methods, a fixed time step of $\Delta t = 1$ fs was utilized to ensure strict phase fidelity over the entire 20 ns (1000 cycles)

computational window. Double-precision floating-point (64-bit) arithmetic was used throughout. The local truncation error for RK4 is $O(\Delta t^5)$, resulting in a cumulative global error of $O(\Delta t^4)$. At $\Delta t = 1$ fs, the phase drift remains below 10^{-7} rad per 1000 cycles, preserving the integrity of the η -algorithm.

To ensure the integrity of the 22.2 aJ switching energy result, we performed a systematic temporal convergence study. The simulation was benchmarked at three different time steps Δt . The reliability of the results was verified through a convergence study, which demonstrated that while our standard step size was 1 fs, reducing it to 0.5 fs yielded a deviation in the final occupancy of less than 0.01%, confirming that the solution has reached the convergence plateau. The 1.0 fs step size effectively suppresses numerical phase drift and non-physical dissipation, providing a robust baseline for the η -algorithm's performance.

To ensure the reliability of the reported results, we performed a multi-parameter error verification. The state occupancy $\langle |v|^2 \rangle$ was monitored throughout 1000 computational cycles. Numerical 'leakage' into higher-order Fock states ($n > 1$) was found to be suppressed below 10^{-6} due to the nonlinear regulator $U = 0.5$. Doubling the basis size from $N = 100$ to $N = 200$ yielded no measurable change in fidelity, confirming the robustness of the truncated Hilbert space approximation.

The computational fidelity was maintained at $> 99.9\%$ across the entire 20-ns window. The primary source of numerical error was the local truncation error of the RK4 integrator, which was strictly controlled by the 1-fs time step, resulting in a cumulative phase drift of less than 10^{-4} rad.

As detailed in the Monte-Carlo verification, these error margins remain stable even under $\pm 5\%$ variations of the physical parameters (η , U , α). The yield was calculated based on a Gaussian distribution of anisotropy field H_k with $\sigma = 5\%$.

Averaging Window: Each data point on the curve was averaged over a steady-state window of 400 ps after an initial relaxation period of 600 ps to eliminate transient pulse artifacts.

3. Initial Conditions and Excitation Pulse

The system was initialized in a near-vacuum state with a residual occupancy of $|v|^2 = 0.01$, representing quantum zero-point fluctuations.

The parametric pump $\mathcal{G}(t)$ was modeled as a sinusoidal pulse at 2Ω (100 GHz parametric excitation) with varying intensity $\eta \in [0.01, 1.5]$. This simulates the sub-picosecond laser or spin-torque excitation pulse described in the η -algorithm.

4. Thermodynamic Consistency

For each operational point at $\langle |v|^2 \rangle \approx 1$, the energy dissipation was calculated as $E = \hbar\Omega_0 \langle |v|^2 \rangle$. At $\Omega = 50$ GHz, this yielded a switching energy of 22.2 aJ, consistent with the theoretical prediction.

To define the operational boundaries of the η -algorithm, we performed a parametric stress-test. Our results identify a Stability Window within which the 22.2 aJ switching remains deterministic. The system tolerates up to a +20% fluctuation in drive intensity η , beyond which the nonlinear regulator U can no longer suppress parametric divergence. The anharmonicity U exhibits the highest tolerance, maintaining logical integrity even with a -75% deviation from the nominal value, provided that it stays above the critical threshold of $U_{\text{crit}} \approx 0.12$. Effective cooling α can vary by an order of magnitude without affecting the switching energy, although values below 10^{-3} significantly limit the coherent lifetime due to accumulative heating.

To evaluate the impact of fabrication variability, we performed a statistical analysis of parameter dispersion across a large-scale cell array. In real-world YIG:Co nanostructures, local variations in film thickness and doping concentration can lead to a distribution of the nonlinear regulator U and damping α .

Our results demonstrate that the η -algorithm is remarkably resilient to these fluctuations. For a standard manufacturing tolerance of $\pm 5\%$, the switching energy remains tightly clustered around the 22.2 aJ target with a computational yield of 99.7%. The nonlinear regulator U acts as a self-correcting mechanism: a local decrease in U is compensated by a slight shift in the stationary occupancy, preserving the logical state. This high tolerance to dispersion confirms that our attojoule architecture is compatible with current industrial lithography and epitaxial growth standards.

Supplement 3: Methodology for Comparative Thermal and Performance Analysis

The primary objective of this analysis is to evaluate the practical scalability of the proposed magnon architecture by benchmarking its computational density against state-of-the-art CMOS standards (e.g., NVIDIA H100 and mobile ARM processors) under identical thermal constraints.

Thermal Modeling Parameters: Ambient Temperature (T_{amb}): 293.15 K (20°C). Target Operational Temperature: 309.75 K (36.6°C) - corresponding to human body temperature. Cooling Conditions: Passive convection and conduction through the printed circuit board (PCB) with an effective heat transfer coefficient of $h_{\text{eff}} = 150 \text{ W}/(\text{m}^2 \text{ K})$. Chip Area (Area): 100 mm^2 (the standard footprint of a modern System-on-Chip, SoC). Thermal Budget (P_{limit}): Defined as the maximum power the system can dissipate passively without exceeding the target temperature: $P_{\text{limit}} = h_{\text{eff}} \text{Area} (T_{\text{target}} - T_{\text{amb}}) \approx 0.25 \text{ W}$

The total computational throughput (operations per second) is derived as follows: $\text{OPS} = P_{\text{limit}} / E_{\text{op}}$

Where the following energy-per-operation (E_{op}) values are utilized: Magnon Logic (this work): $E_{\text{op}} = 22.2 \times 10^{-18} \text{ J}$ (direct switching energy of the localized magnon). Energy-Efficient CMOS: $E_{\text{op}} = 100 \times 10^{-15} \text{ J}$ (optimized mobile logic energy per operation, including parasitic losses from local interconnects).

To ensure a system-level verification, we include the overhead of phase-coherent readout. Based on the thermal noise limit at 300K and a conservative 1% detector efficiency, the readout energy is estimated at 0.41 aJ per bit. The combined energy for a "Switch-and-Read" cycle is: $E_{\text{total}} = 22.61 \text{ aJ}$. Under a passive thermal budget of 0.25 W (defined by the 36.6°C limit), the chip supports an aggregate throughput of: $\text{OPS} \approx 1.1 \times 10^{16}$. This allows for the simultaneous operation of approximately 110,000 cells at 100 GHz or the entire 10-billion cell array at a 1.1 MHz effective duty cycle, surpassing intrinsic core CMOS performance by 4,500 times under identical cooling constraints without consideration of readout and interconnect architectures.

Supplement 4. Derivation of the master equation and its semiclassical (Bogoliubov) reduction

This supplement makes explicit the connection, requested during review, between the Hamiltonian (Eq. 4), the open-system (Lindblad) master equation, and the semiclassical $u(t)$, $v(t)$ equations of motion used for the numerical results throughout Sections 5-6 and Fig. 3. We distinguish explicitly between steps that are exact/standard and steps that involve a controlled approximation, following the Established / Approximation / Conjectural labeling used elsewhere in this work.

S4.1 Master equation [Established]

The open-system dynamics of the mode operator $\hat{A}$ are governed by the Lindblad master equation for the density matrix ρ :

$$d\rho/dt = -i[\hat{H}(t), \rho] + \kappa (\hat{A}\rho\hat{A}^\dagger - \frac{1}{2}\{\hat{A}^\dagger\hat{A}, \rho\})$$

with $\hat{H}(t)$ given by Eq. (4) ($\hbar = 1$, as elsewhere in this work) and a single dissipation channel with rate $\kappa = 2\alpha\Omega(t)$, fixed by the Gilbert damping constant α so that the resulting linear amplitude decay rate is exactly $\alpha\Omega(t)$ (see S4.4).

S4.2 Linear ($U = 0$) limit: exact correspondence with Eq. (2)-(3) [Established]

For a single Lindblad channel with jump operator $\hat{A}$, the exact Heisenberg-Langevin equation of motion is $d\hat{A}/dt = -i[\hat{A}, \hat{H}(t)] - (\kappa/2)\hat{A} + \sqrt{\kappa}\hat{A}_{\text{in}}(t)$, where $\hat{A}_{\text{in}}(t)$ is vacuum/thermal input noise. In the linear limit ($U = 0$), $\hat{H}$ reduces to the standard single-mode parametric (degenerate) oscillator Hamiltonian $\Omega(t)\hat{A}^\dagger\hat{A} + \mathcal{G}(t)(\hat{A}^{\dagger 2} + \hat{A}^2)$, whose Bogoliubov (positive-/negative-frequency) decomposition is exactly the linear system already given in Eq. (2)-(3) of the main text. No new derivation is required here: the u, v mixing via $\dot{\Omega}/2\Omega$ used throughout this work is the standard, exact consequence of this quadratic Hamiltonian.

S4.3 Nonlinear term: mean-field (Hartree) reduction [Approximation]

The quartic term $U(\hat{A}^\dagger\hat{A})^2$ does not preserve the simple linear structure of S4.2: evaluating $[\hat{A}, \hat{H}]$ gives a term proportional to $\hat{A}^\dagger\hat{A}^2$, coupling the equation of motion to a higher-order moment rather than closing on u, v alone. We close this hierarchy at the mean-field (Hartree) level, $\langle\hat{A}^\dagger\hat{A}^2\rangle \approx \langle\hat{A}^\dagger\rangle\langle\hat{A}\rangle^2$, which is the standard approximation used for Kerr-type nonlinearities in this

context. In the Bogoliubov language of Eq. (2)-(3), this closure is what generates the state-dependent frequency shift $\Omega \rightarrow \Omega + U\Omega|v|^2$ used in the main text: the nonlinear detuning is not an independent postulate but the Hartree reduction of the $U(\hat{A}^\dagger \hat{A})^2$ term. This approximation is controlled by the relative size of number fluctuations: it is reliable when the fluctuation in $|v|^2$ is small compared to its mean, and becomes less reliable otherwise. This gives a second, independent reason — beyond simply being far from the single-magnon target — why the low-damping, $U = 0.12$ regime characterized in Fig. 3 (relative fluctuation $\approx 50\%$) should be treated with caution: the same large fluctuation that makes the occupancy non-deterministic also weakens the mean-field approximation used to derive the equation of motion in the first place. The calibrated regime ($U \approx 0.5$, $\alpha \approx 0.01$, relative fluctuation $\approx 14\%$) is correspondingly better justified on both counts.

S4.4 Damping: from the single-mode dissipator to the (u , v) equations [Approximation]

For the single mean amplitude $\langle \hat{A} \rangle$ alone, the Lindblad dissipator of S4.1 gives an exact linear decay rate $\kappa/2 = \alpha \Omega(t)$, which is the origin of the Gilbert-damping identification $\kappa = 2 \alpha \Omega(t)$ used throughout this work. Extending this damping to each of the two Bogoliubov components u , v of the coupled (S4.2) system is done here by natural symmetry — applying the same rate $\alpha \Omega(t)$ to both components — rather than by an independent re-derivation of the Lindblad equation in the explicitly time-dependent quasiparticle (u , v) basis. A fully rigorous treatment would derive the master equation directly in this time-dependent basis, accounting for the fact that the basis transformation itself is time-dependent; this is left for future, more mathematically-focused work.

S4.5 Summary

The linear mixing structure of the u , v equations (Eq. 2-3) is an exact, standard consequence of the master equation in S4.1. The nonlinear detuning term and the damping term applied to each Bogoliubov component are both physically motivated, standard approximations (Hartree closure; symmetric damping extension) rather than independent operator-level derivations from Eq. (4). This does not invalidate the numerical results in Sections 5-6, which remain internally consistent and, for the occupancy statistics, are further validated against the independent full 100-level Fock-basis Lindblad simulation (Figs. 1-2); it does mean that the quantitative reliability of the semiclassical reduction is itself tied to the fluctuation level reported in Fig. 3, as noted in S4.3.